

NOT, AND, OR

Name: _____ Date: _____

Patience rule: When a problem asks for *all* assignments, list every row (8 for three variables, 16 for four). Show the compound evaluated step by step for each row you claim works.

Deduction problems — Connectives

D2.1 DEDUCTION

Let p, q, r be unknown truth values. Find **every** assignment (T/F to each letter) that makes the following compound **true**:

$$(p \vee \neg q) \wedge (\neg p \vee r) \wedge (q \vee \neg r) \wedge \neg(p \wedge q \wedge r)$$

1. Build an 8-row table with columns for p, q, r , and each of the four conjuncts, plus the final AND.
2. List every row where the final column is T.
3. Explain in one paragraph why any row you did *not* list fails (refer to which conjunct breaks).

FULL 8-ROW TABLE + LIST + EXPLANATION

D2.2 DEDUCTION

A club's membership rule in English:

“You may enter the lounge **exactly when** you are a paid member **and** either you have a guest pass **or** it is not after 9 p.m.”

Define m = paid member, g = has guest pass, l = it is after 9 p.m., e = may enter the lounge.

1. Translate as $e \equiv m \wedge (g \vee \neg l)$ using only $\neg, \wedge, \vee$ — write both sides and explain why they must match (entry allowed $\leftrightarrow$ condition met).
2. Evaluate for **all eight** assignments of (m, g, l) : compute the right-hand condition and decide whether e must be T or F in each row.
3. Which of the eight worlds allow entry (e true)? Which forbid it? Match at least two rows to a short English story (“Member without pass before 9...”).

SYMBOLIZATION + 8-ROW EVALUATION + STORIES

D2.3 DEDUCTION

Are these two compounds **logically equivalent** (same truth value in every row)? You may not appeal to De Morgan by name — prove it by exhaustive check or produce one counterexample row.

$$A: \neg(p \wedge (\neg q \vee r)) \quad B: (\neg p \vee q) \wedge (\neg p \vee \neg r)$$

1. Build an 8-row table with sub-columns for the inner parts of A and B.
2. State “equivalent” or “not equivalent.”
3. If not equivalent, identify every row where they differ. If equivalent, confirm all eight rows match.

8-ROW COMPARISON TABLE + VERDICT

D2.4 DEDUCTION

Four switches p, q, r, s control a warning light. The light is ON exactly when:

$$(p \wedge q) \vee (\neg p \wedge r \wedge \neg s) \vee (q \wedge \neg r \wedge s)$$

1. How many rows does a complete truth table need? Build it — all rows, all sub-columns for the three disjuncts, final OR.
2. Count how many assignments turn the light ON.
3. The engineer claims: “If p and r are both ON, the light must be ON.” Test the claim by checking **every** row where $p = \text{T}$ and $r = \text{T}$. Is the engineer right?

16-ROW TABLE (ABBREVIATE ONLY IF EVERY ROW IS STILL VISIBLE) + COUNTS + ENGINEER CHECK

